

Milk-induced pulmonary disease in infants (Heiner syndrome).

Heiner syndrome (HS) is a food hypersensitivity pulmonary disease that affects primarily infants, and is mostly caused by cow's milk (CM). Only a few reports have been published, which may be due to its misdiagnosis. We review here a series of eight cases. When first diagnosed they were 4-29 months of age. They were fed CM from birth and their chronic respiratory symptoms began at age 1-9 months. The symptoms were in the form of cough in seven, wheezing in three, hemoptysis in two, nasal congestion in three, dyspnea in one, recurrent otitis media (OM) in three, recurrent fever in four, anorexia, vomiting, colic or diarrhea in five, hematochezia in one, and failure to thrive (FTT) in two. All had radiologic evidence of pulmonary infiltrates. High titers of precipitating antibodies to CM proteins were demonstrated in six of six and milk-specific immunoglobulin E (IgE) was positive in one of two. Pulmonary hemosiderosis (PH) was confirmed in one patient who showed iron-laden macrophages (ILM) in the bronchoalveolar lavage (BAL), gastric washing, and open lung biopsy. Additional findings, in a descending frequency, were eosinophilia, anemia, and elevated level of total IgM, IgE or IgA. Milk elimination resulted in remarkable improvement in symptoms within days and clearing of the pulmonary infiltrate within weeks. Parents consented to milk challenge in only three cases, all of whom developed recurrence of symptoms. After 2 yr of milk avoidance in one patient, milk challenge was tolerated for 2 months, and then the patient developed symptoms, serum milk precipitins, pulmonary infiltrate, and ILM. The HS should be suspected in young children with chronic pulmonary disease of obscure cause. The diagnosis is supported with a positive milk precipitin test and improvement on a trial of milk elimination. Severe cases may be complicated with PH, which should be suspected in the presence of anemia or hemoptysis and be confirmed with the demonstration of ILM.